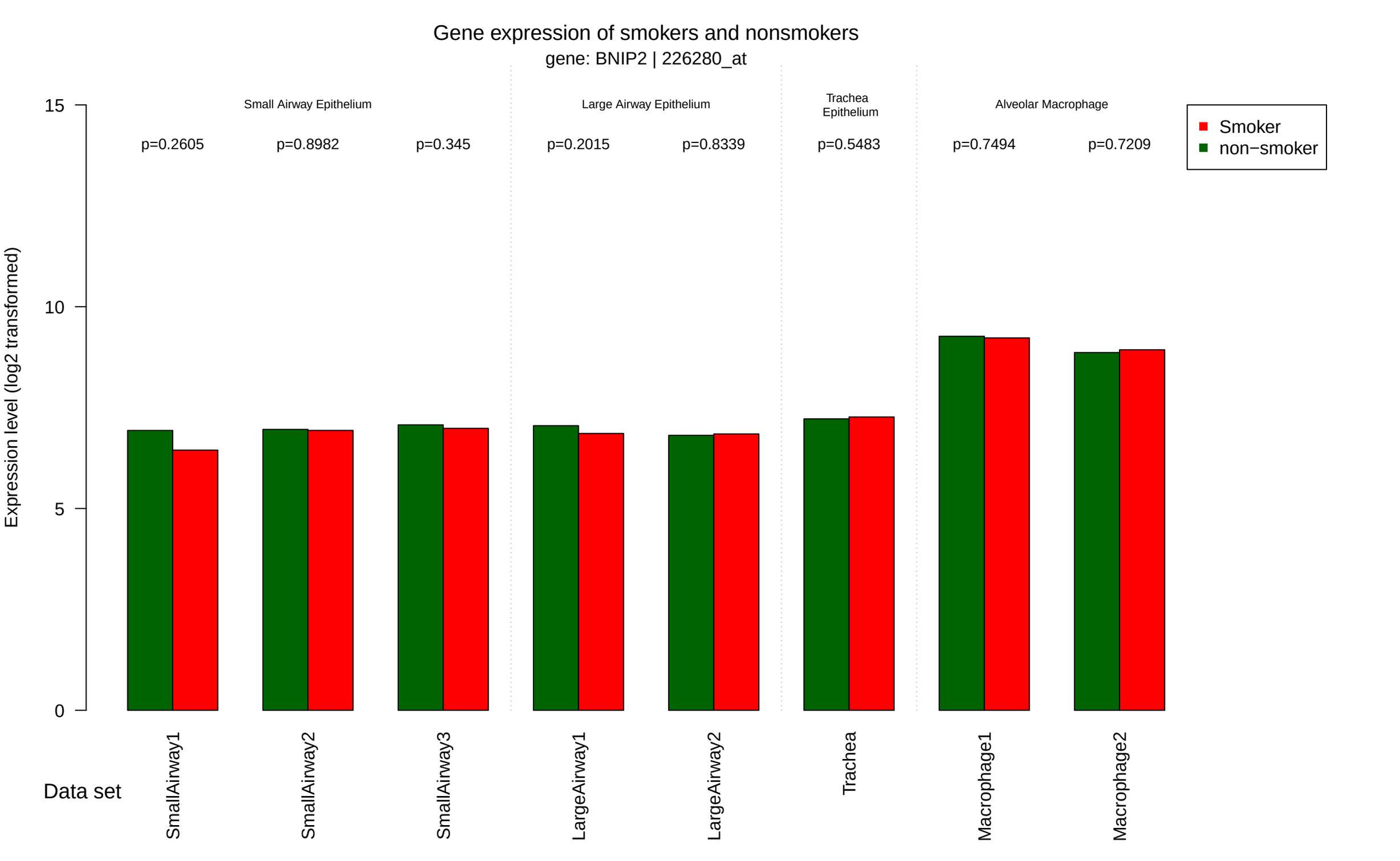
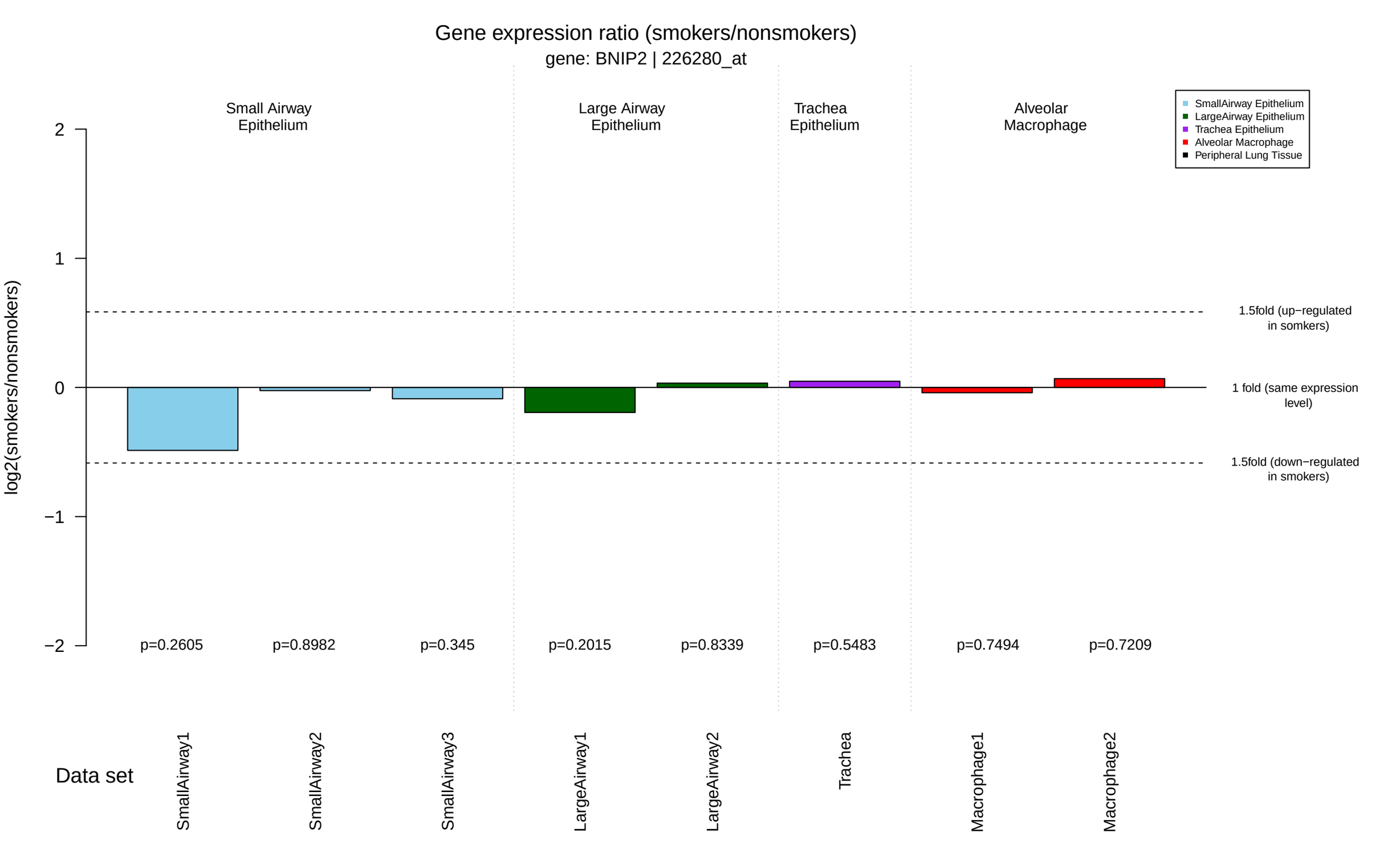
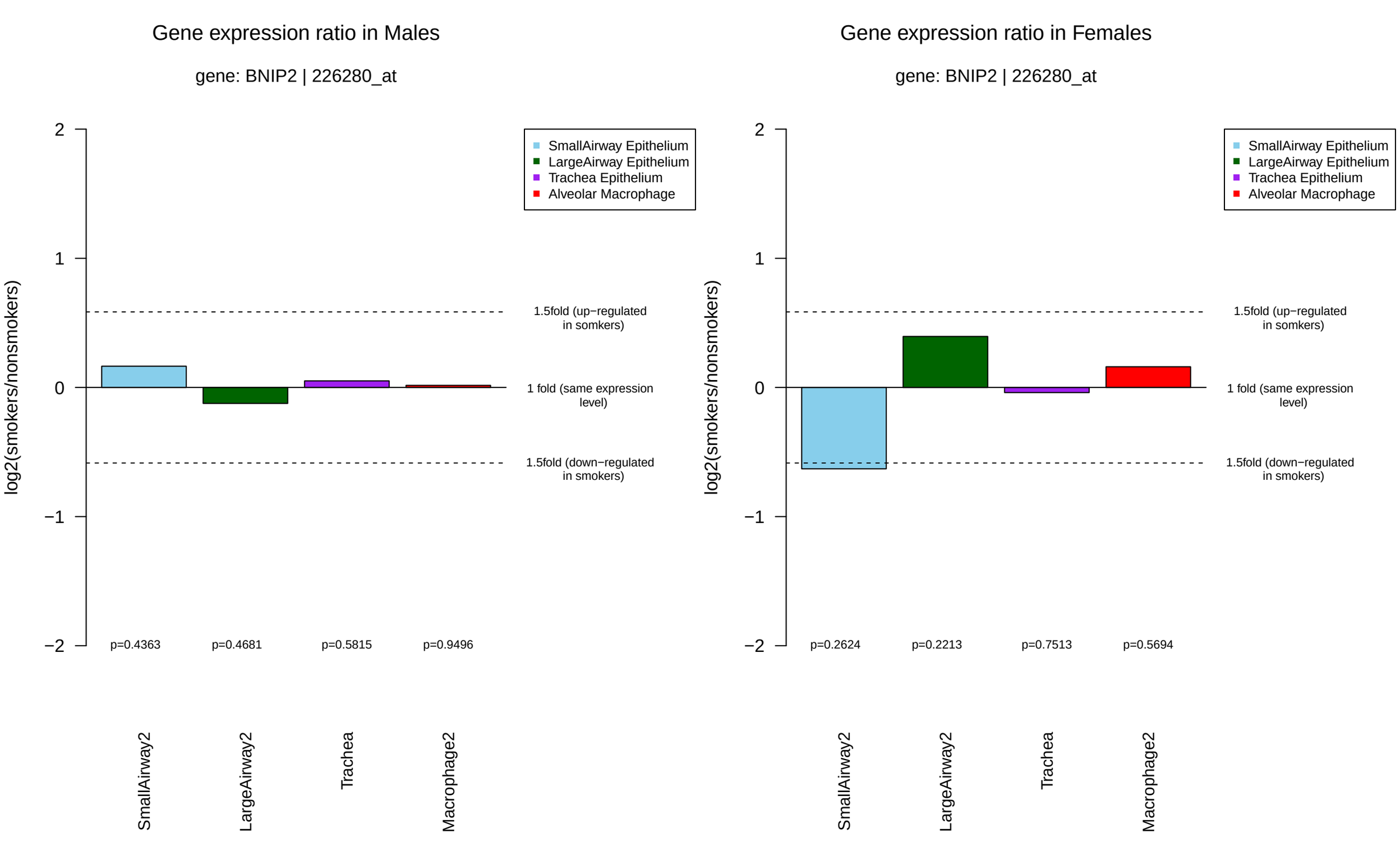
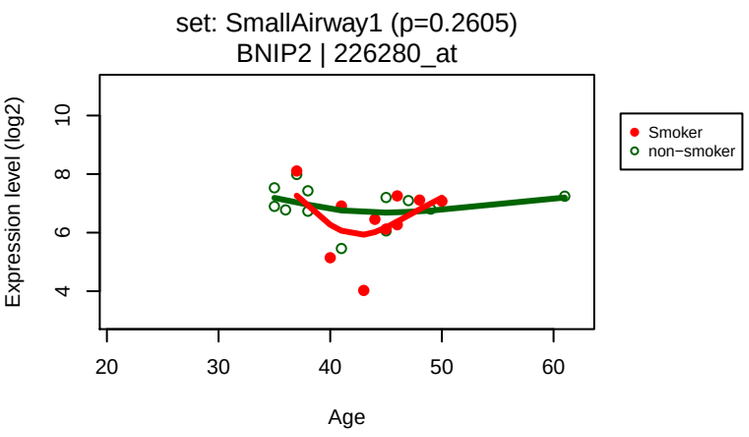
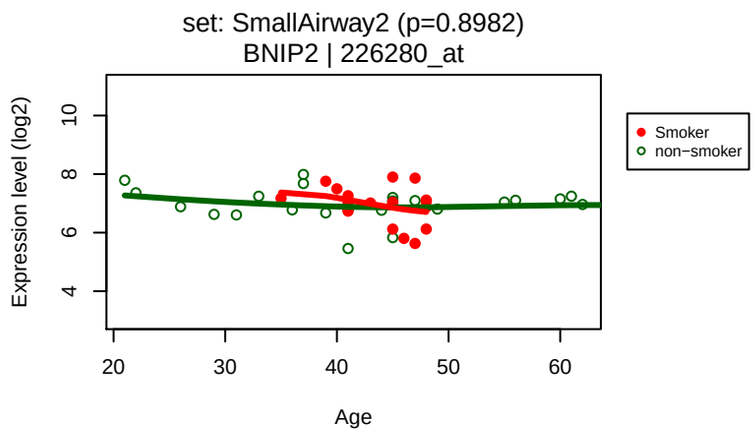
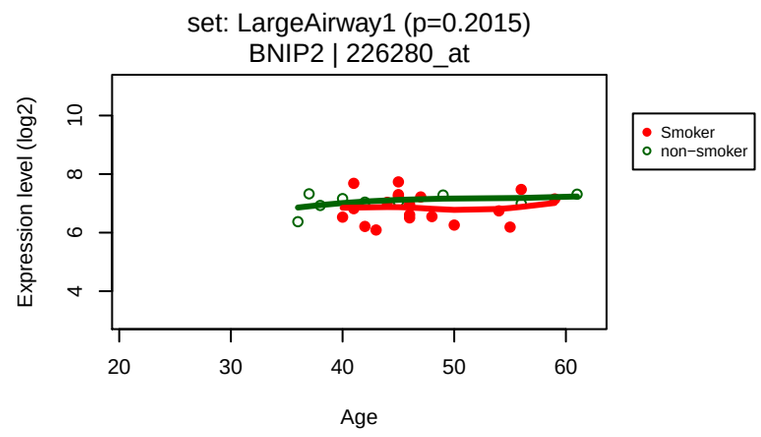

Gene expression of smokers and nonsmokers
gene: BNIP2 | 226280_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2605
p=0.8982
p=0.345
p=0.2015
p=0.8339
p=0.5483
p=0.7494
p=0.7209
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

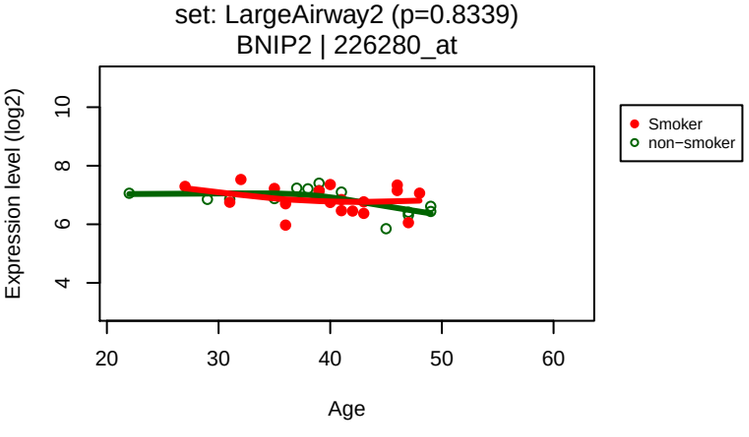

Gene expression ratio (smokers/nonsmokers)
gene: BNIP2 | 226280_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2605
p=0.8982
p=0.345
p=0.2015
p=0.8339
p=0.5483
p=0.7494
p=0.7209
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

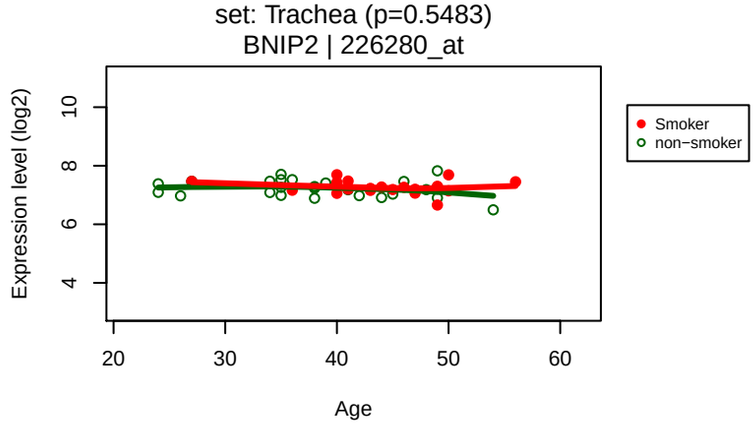

Gene expression ratio in Males
gene: BNIP2 | 226280_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4363
p=0.4681
p=0.5815
p=0.9496
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BNIP2 | 226280_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2624
p=0.2213
p=0.7513
p=0.5694
SmallAirway2
LargeAirway2
Trachea
Macrophage2

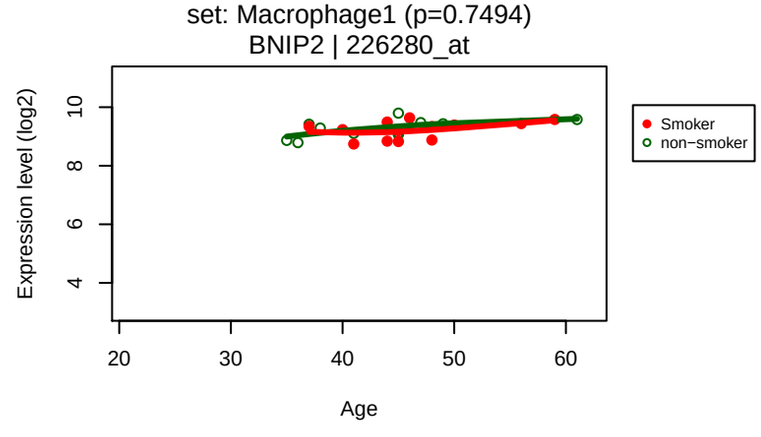

set: SmallAirway1 (p=0.2605)
BNIP2 | 226280_at
Expression level (log2)
Age
Smoker
non−smoker

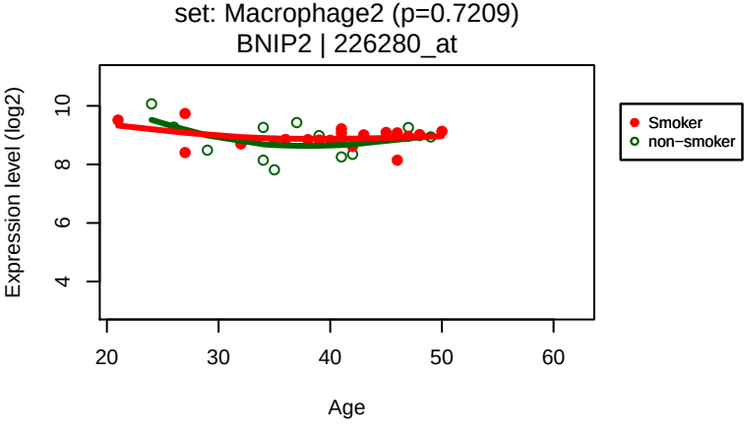

set: SmallAirway2 (p=0.8982)
BNIP2 | 226280_at
Expression level (log2)
Age
Smoker
non−smoker

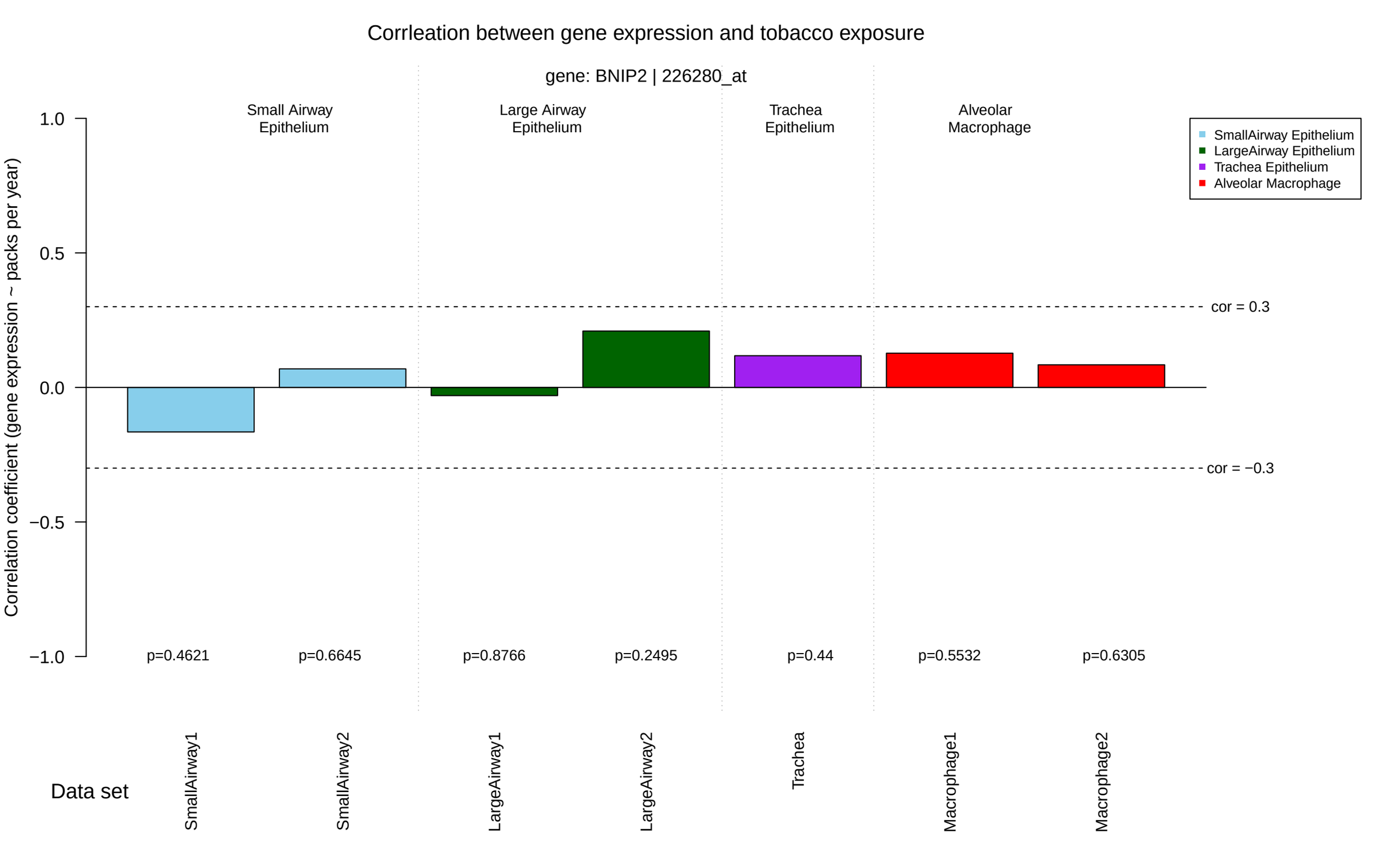

set: LargeAirway1 (p=0.2015)
BNIP2 | 226280_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8339)
BNIP2 | 226280_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5483)
BNIP2 | 226280_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7494)
BNIP2 | 226280_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7209)
BNIP2 | 226280_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BNIP2 | 226280_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4621
p=0.6645
p=0.8766
p=0.2495
p=0.44
p=0.5532
p=0.6305
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2